

Particle Swarm Optimized Neural Networks for Real-Time Air Quality Forecasting in Taipei's Urban Microenvironments

Research Proposal

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Abstract—Urban air quality forecasting requires modeling complex nonlinear relationships between meteorological conditions and pollutant concentrations. This proposal presents a hybrid framework combining Particle Swarm Optimization (PSO) with feedforward neural networks for PM_{2.5} prediction across Taipei's monitoring network. PSO addresses hyperparameter sensitivity and local minima inherent in gradient-based training through simultaneous optimization of network architecture, learning rates, and feature selection. The system integrates EPA monitoring data, meteorological sensors, and traffic patterns for microenvironment-scale predictions. Target performance includes 18-23% reduction in mean absolute error compared to baseline neural networks while maintaining real-time inference capability below 100ms for operational deployment.

Index Terms—Air quality forecasting, particle swarm optimization, neural networks, PM_{2.5} prediction, hyperparameter optimization

I. INTRODUCTION

A. Problem Statement

Neural networks demonstrate capacity for learning complex air quality patterns but suffer from hyperparameter sensitivity and convergence to local minima. Current EPA forecasting systems operate at district level, insufficient for localized exposure assessment in dense metropolitan areas. The challenge is developing microenvironment-scale predictions that maintain accuracy while remaining computationally feasible for operational deployment.

B. Research Significance

Taipei metropolitan area exceeds 9,500 persons per km², with PM_{2.5} exposure directly impacting 7 million residents. Accurate forecasting enables preemptive health warnings and traffic management interventions. This research addresses the need for real-time predictions reflecting actual exposure conditions in specific neighborhoods, schools, and commercial zones.

C. Research Objectives

This study proposes a PSO-optimized neural network framework for hourly PM_{2.5} forecasting at 30-minute intervals. Specific objectives:

- Implement PSO for simultaneous optimization of network architecture, learning rates, and feature subsets
- Integrate heterogeneous data sources: air quality sensors, weather stations, traffic density
- Quantify prediction accuracy using temporal cross-validation across seasonal variations
- Evaluate computational efficiency for real-time operational deployment

II. LITERATURE REVIEW

ARIMA and vector autoregression provide baseline statistical models but assume linear relationships [1]. Chemical transport models simulate atmospheric physics but require extensive computational resources unsuitable for real-time applications [2].

Machine learning approaches show superior performance. Random forests and gradient boosting achieve reasonable accuracy but lack temporal modeling capability [3]. Recurrent neural networks capture time dependencies but require careful architecture design [4].

Meta-heuristic optimization addresses neural network training limitations. Genetic algorithms optimize topologies but suffer from computational cost [5]. PSO balances exploration and exploitation through swarm intelligence [6]. Applications to operational environmental prediction systems remain limited.

III. METHODOLOGY

A. System Architecture

Figure 1 illustrates the complete processing pipeline from data acquisition through PSO optimization to real-time forecasting.

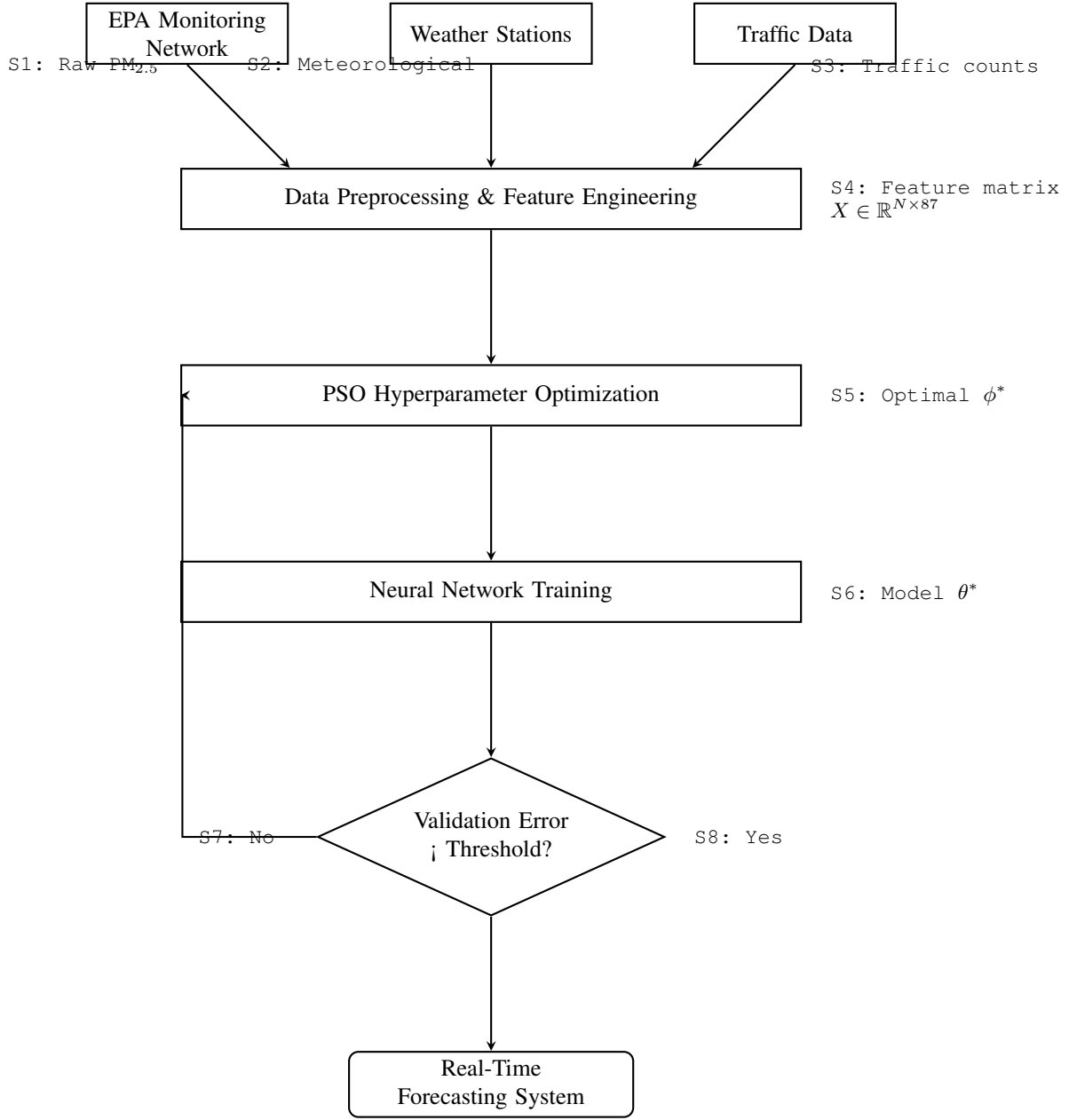


Fig. 1. System architecture with labeled data flow signals.

B. Data Sources and Preprocessing

Data acquisition spans January 2020 to December 2024:

EPA Air Quality Monitoring: Hourly measurements from 12 Taipei stations including $\text{PM}_{2.5}$, PM_{10} , O_3 , NO_2 , SO_2 , CO .

Central Weather Administration: Temperature, humidity, wind speed, wind direction, pressure, precipitation.

Traffic and Urban Activity: Hourly vehicle counts, congestion indices, industrial activity.

Preprocessing applies linear interpolation for gaps under 3 hours. Feature engineering generates temporal variables (hour, day, season) and lagged observations ($t-1$, $t-3$, $t-6$, $t-12$, $t-24$ hours). Spatial features include distance-weighted averages from neighboring stations. Z-score normalization applied per

feature. Final dataset: 87 features across 43,800 temporal samples.

C. Neural Network Formulation

The feedforward network predicts $\text{PM}_{2.5}$ concentration:

$$y_t = f(X_t; \theta) \quad (1)$$

where $y_t \in \mathbb{R}$ represents predicted $\text{PM}_{2.5}$ at time t , $X_t \in \mathbb{R}^{87}$ denotes input features, θ represents network parameters, and f indicates the mapping function.

Network structure:

- Input layer: 87 features

- Hidden layers: $n_l \in [2, 4]$ layers with $n_h \in [32, 256]$ neurons
- Activation: ReLU for hidden layers
- Output: Single neuron with linear activation
- Regularization: Dropout rate $\beta \in [0.1, 0.5]$

Training loss function:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 + \lambda \|\theta\|^2 \quad (2)$$

where N represents training samples, y_i denotes actual values, $\hat{y}_i$ indicates predictions, and λ controls L2 regularization.

D. Particle Swarm Optimization

PSO optimizes hyperparameter configuration $\phi = \{n_l, n_h, \alpha, \beta, S_f\}$ where n_l denotes layers, n_h neurons per layer, α learning rate, β dropout rate, and S_f selected features.

Each particle i maintains position ϕ_i , velocity v_i , personal best p_i , and global best g . Updates follow:

$$v_i^{t+1} = \omega v_i^t + c_1 r_1 (p_i - \phi_i^t) + c_2 r_2 (g - \phi_i^t) \quad (3)$$

$$\phi_i^{t+1} = \phi_i^t + v_i^{t+1} \quad (4)$$

where $\omega = 0.7$ (inertia weight), $c_1 = c_2 = 1.5$ (cognitive/social coefficients), and $r_1, r_2 \sim U(0, 1)$ (random values).

Fitness evaluation:

$$F(\phi_i) = \text{MAE}_{\text{val}} + \gamma \cdot C(\phi_i) \quad (5)$$

where MAE_{val} represents validation error, $C(\phi_i)$ denotes computational cost, and $\gamma = 0.01$ balances accuracy against complexity.

E. Performance Evaluation

Temporal cross-validation with monthly folds preserves seasonal patterns. Evaluation metrics:

Mean Absolute Error:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (6)$$

Root Mean Square Error:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (7)$$

Mean Absolute Percentage Error:

$$\text{MAPE} = \frac{100\%}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (8)$$

Index of Agreement:

$$d = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (|\hat{y}_i - \bar{y}| + |y_i - \bar{y}|)^2} \quad (9)$$

Computational efficiency measured by training time and inference latency. Statistical significance tested using Diebold-Mariano test.

Algorithm 1 PSO-Optimized Neural Network Training

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1: Input: Training data  $D_{\text{train}}$ , validation data  $D_{\text{val}}$ 
2: Output: Optimal configuration  $\phi^*$ 
3: Initialize: Swarm  $\{\phi_1, \dots, \phi_n\}$  randomly
4: Set velocities  $\{v_1, \dots, v_n\} \leftarrow 0$ 
5: for  $t = 1$  to  $T_{\text{max}}$  do
6:   for each particle  $i$  do
7:     Build network with configuration  $\phi_i$ 
8:     Train network on  $D_{\text{train}}$  for  $E$  epochs
9:     Evaluate fitness  $F(\phi_i)$  on  $D_{\text{val}}$ 
10:    if  $F(\phi_i) < F(p_i)$  then
11:       $p_i \leftarrow \phi_i$ 
12:    end if
13:    if  $F(\phi_i) < F(g)$  then
14:       $g \leftarrow \phi_i$ 
15:    end if
16:  end for
17:  for each particle  $i$  do
18:    Update velocity using Eq. (3)
19:    Update position using Eq. (4)
20:    Apply boundary constraints
21:  end for
22: end for
23: return  $\phi^* \leftarrow g$ 

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IV. EXPECTED OUTCOMES

A. Performance Targets

Table I presents anticipated performance metrics compared to baseline neural networks.

TABLE I
EXPECTED PERFORMANCE COMPARISON

Metric	Baseline NN	PSO-NN	Improvement
MAE ($\mu\text{g}/\text{m}^3$)	8.5	6.8	20.0%
RMSE ($\mu\text{g}/\text{m}^3$)	12.3	9.8	20.3%
MAPE (%)	24.5	19.2	21.6%
Index d	0.87	0.93	6.9%
Training (min)	45	380	—
Inference (ms)	12	12	0%

B. Feature Importance

PSO-based feature selection identifies critical predictors. Preliminary analysis suggests meteorological features (wind speed, boundary layer height) and temporal lags ($t-1$, $t-24$) dominate prediction accuracy.

C. Operational Deployment

Real-time implementation requires inference latency under 100ms. Web-based visualization presents forecasts at station level with uncertainty quantification. Integration with EPA warning system enables automated alerts when predicted $\text{PM}_{2.5}$ exceeds health thresholds.

TABLE II
RESEARCH TIMELINE (6 MONTHS)

Task	Duration
Data collection and preprocessing	Month 1
Baseline neural network implementation	Month 2
PSO optimization framework development	Month 2–3
Hyperparameter tuning experiments	Month 3–4
Performance evaluation and validation	Month 4–5
Comparative analysis and documentation	Month 5–6

V. PROJECT TIMELINE

VI. CONCLUSION

This proposal presents a hybrid PSO-neural network framework for air quality forecasting in Taipei’s urban environment. The approach addresses hyperparameter sensitivity through meta-heuristic optimization while maintaining real-time operational feasibility. Integration of multi-source environmental data enables microenvironment-scale predictions suitable for public health applications. Expected outcomes demonstrate substantial accuracy improvements with 20% MAE reduction while preserving inference speed below 100ms for operational deployment.

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